



Fundamentals of Optics

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Room 239, South Tower, College of Engineering

Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

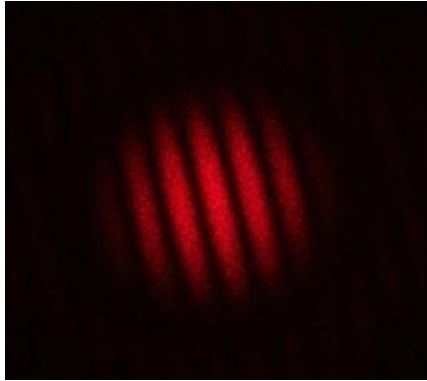
Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

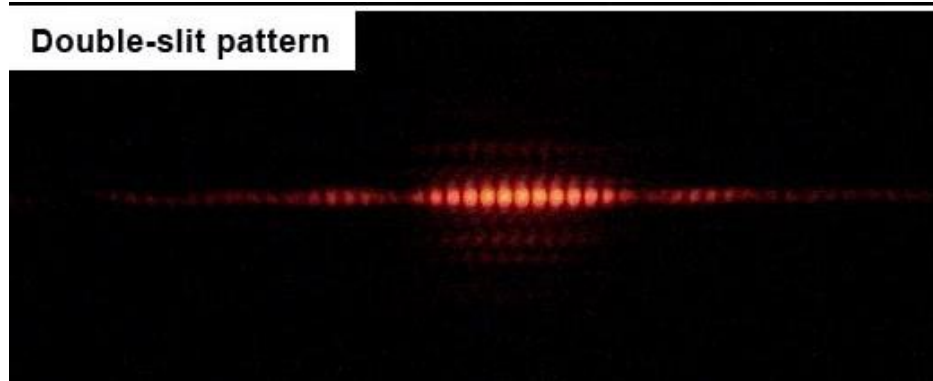
CHAPTER 6

THE SUPERPOSITION OF WAVES

Study the phenomena of **interference**, **diffraction** and **polarization**



Interference of light through two narrow holes



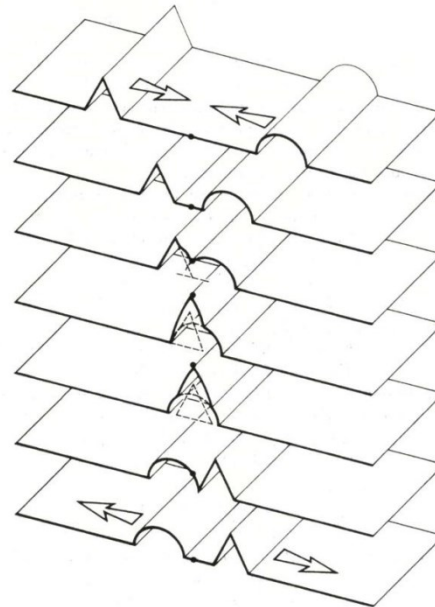
Multi-Slit Diffraction Experiments



Rainbow effect from cross polarization of clear plastic objects.

Share a common conceptual basis:
lightwave overlap

The resultant disturbance at any point in a medium is the algebraic sum of the separate constituent waves in **linear** systems.



Content

- ☐ The Addition of Waves of the Same Frequency
- ☐ The Addition of Waves of Different Frequency
- ☐ A harmonic Periodic Waves
- ☐ Nonperiodic Waves

重点

The addition of Waves of the Same Frequency

The algebraic method

An harmonic disturbance $E(x, t) = E_0 \sin[\omega t - (kx + \epsilon)]$

E_0 : the amplitude of the harmonic disturbance propagating along the positive x-direction.

Separate the space and time parts of the phase

$$E(x, t) = E_0 \sin(\omega t + \alpha) \quad \text{and} \quad \alpha(x, \epsilon) = -(kx + \epsilon)$$

Suppose that there are two waves with the same frequency and speed:

$$E_1 = E_{01} \sin(\omega t + \alpha_1) \quad \text{and} \quad E_2 = E_{02} \sin(\omega t + \alpha_2)$$

$$\alpha_1(x_1, \epsilon_1) = -(kx_1 + \epsilon_1) \quad \alpha_2(x_2, \epsilon_2) = -(kx_2 + \epsilon_2)$$

The result disturbance is the linear superposition of these waves:

$$E = E_1 + E_2$$

We can anticipate a sinusoidal function of frequency ω with an amplitude (E_0) and phase (α).

$$\begin{aligned}
 E &= E_1 + E_2 = E_{01}\sin(\omega t + \alpha_1) + E_{02}\sin(\omega t + \alpha_2) \\
 &= E_{01}(\sin\omega t \cos\alpha_1 + \cos\omega t \sin\alpha_1) \\
 &\quad + E_{02}(\sin\omega t \cos\alpha_2 + \cos\omega t \sin\alpha_2) \\
 &= (E_{01}\cos\alpha_1 + E_{02}\cos\alpha_2)\sin\omega t \\
 &\quad + (E_{01}\sin\alpha_1 + E_{02}\sin\alpha_2)\cos\omega t
 \end{aligned}$$

Let $E_0\cos\alpha = E_{01}\cos\alpha_1 + E_{02}\cos\alpha_2$

$$E_0\sin\alpha = E_{01}\sin\alpha_1 + E_{02}\sin\alpha_2$$

$$E_0^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos(\alpha_2 - \alpha_1)$$

$$\tan\alpha = \frac{E_{01}\sin\alpha_1 + E_{02}\sin\alpha_2}{E_{01}\cos\alpha_1 + E_{02}\cos\alpha_2}$$

The total disturbance:

$$E = E_0\cos\alpha\sin\omega t + E_0\sin\alpha\cos\omega t$$

$$E = E_0\sin(\omega t + \alpha)$$

The composite wave is harmonic and of the same frequency as the constituent with different amplitude and phase.

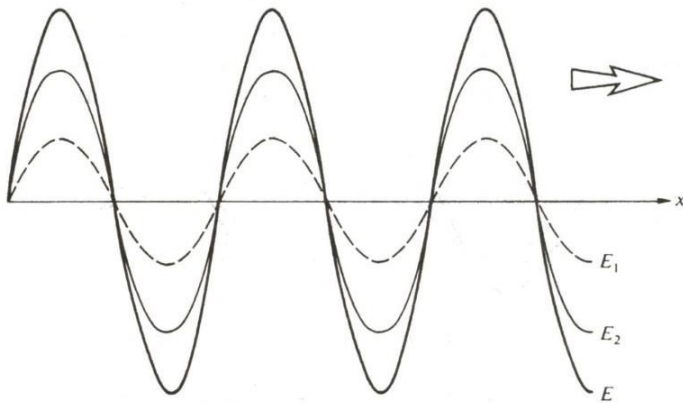
$$E_0^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos(\alpha_2 - \alpha_1)$$

The flux density is proportional to its amplitude squared. But it is not simply the sum of the component of flux densities.

The interference term: $2E_{01}E_{02}\cos(\alpha_2 - \alpha_1)$

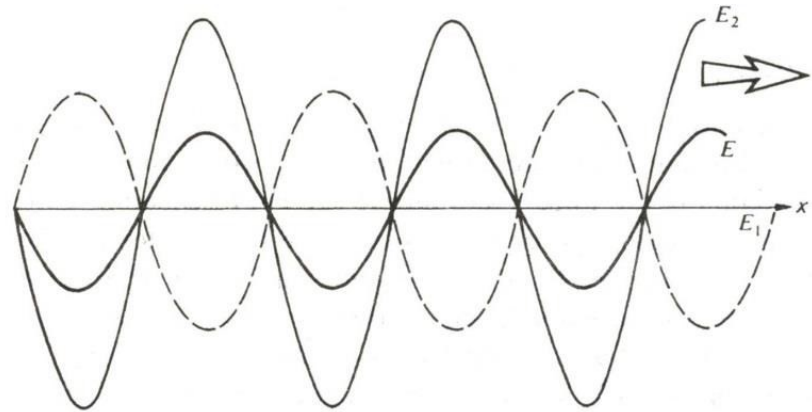
Difference in phase between the two interfering waves: $\delta \equiv \alpha_2 - \alpha_1$

In phase: $\delta = \pm 2n\pi$



Crest overlaps crest

180° out of phase: $\delta = \pm (2n+1)\pi$



Trough overlaps crest

Phase difference:

$$\delta = (kx_1 + \epsilon_1) - (kx_2 + \epsilon_2) = \frac{2\pi}{\lambda} (x_1 - x_2) + (\epsilon_1 - \epsilon_2)$$

x_1 and x_2 : the distances from the sources to the point of observation.

If the waves are initially in-phase: $\delta = \frac{2\pi}{\lambda} (x_1 - x_2)$

In the media, $n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$, $\delta = \frac{2\pi}{\lambda_0} n(x_1 - x_2)$

Optical path difference: the difference in the two optical path lengths (OPL)

$$\Lambda = n(x_1 - x_2)$$

$$\delta = 2\pi \frac{\Lambda}{\lambda_0} = k_0 \Lambda$$

One route is δ radians longer than the other.

Coherent: $\epsilon_1 - \epsilon_2 = \text{constant}$ regardless of its value.

Example

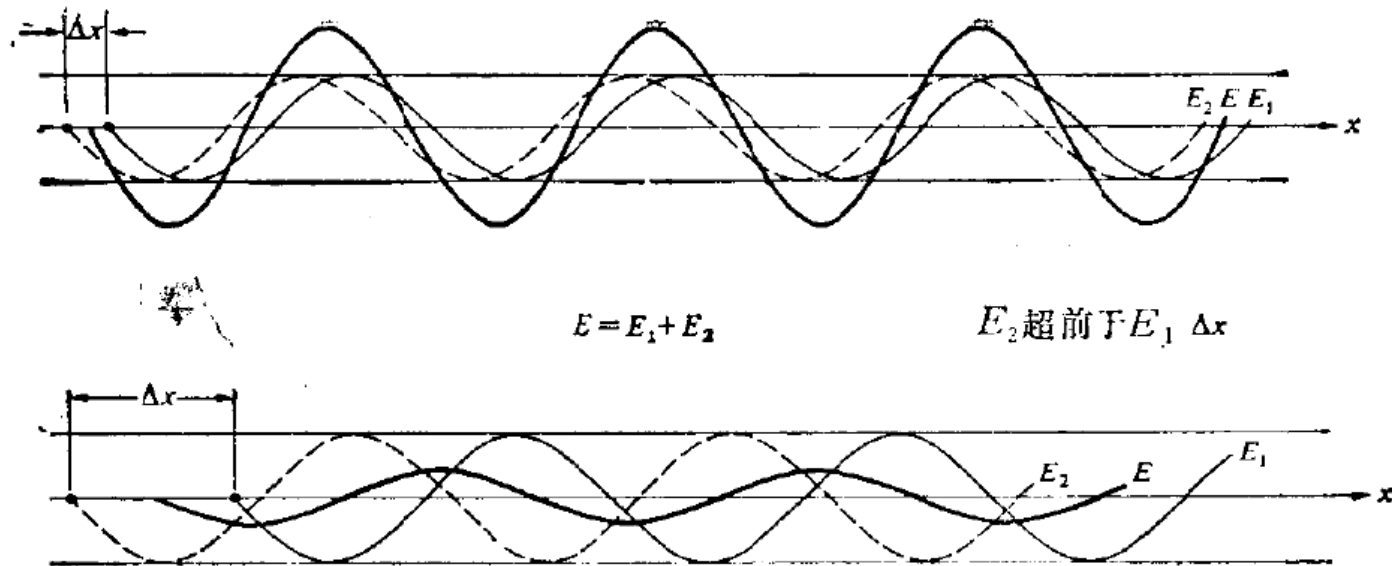
$$E_1 = E_{01} \sin[\omega t - k(x + \Delta x)] \quad E_2 = E_{02} \sin(\omega t - kx)$$

$$\epsilon_1 = \epsilon_2 \quad E_{01} = E_{02} \quad \alpha_2 - \alpha_1 = k\Delta x$$

$$E = 2E_{01} \cos\left(\frac{k\Delta x}{2}\right) \sin\left[\omega t - k\left(x + \frac{\Delta x}{2}\right)\right]$$

Case I: $\Delta x \ll \lambda$ $E_0 \approx 2E_{01}$ **Constructive interference (相长干涉)**

Case II: $\Delta x = \lambda/2$ $E_0 = 0$ **Destructive interference (相消干涉)**



The Superposition of Many Waves

Superposition of any number of coherent harmonic waves having a given frequency and traveling in the same direction leads to a harmonic wave of that same frequency.

The sum of N waves

$$E = \sum_{i=1}^n E_{0i} \cos(\alpha_i \pm \omega t)$$

is given by $E = E_0 \cos(\alpha \pm \omega t)$

where

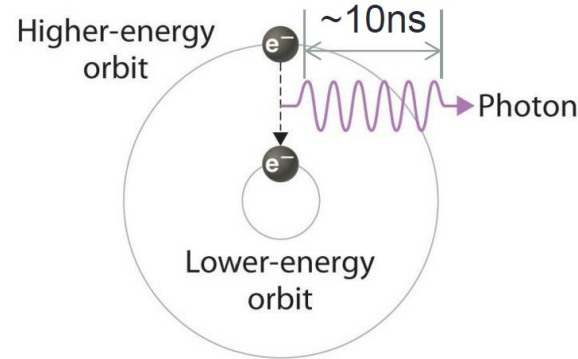
$$E_0^2 = \sum_{i=1}^N E_{0i}^2 + 2 \sum_{j>i}^N \sum_{i=1}^N E_{0i} E_{0j} \cos(\alpha_i - \alpha_j)$$

$$\tan \alpha = \frac{\sum_{i=1}^n E_{0i} \sin \alpha_i}{\sum_{i=1}^n E_{0i} \cos \alpha_i}$$

In the case of in-phase coherent sources ($\alpha_i = \alpha_j$): $E_0^2 = N^2 E_{01}^2$

Example: atomic emitters (N sources)

Imagine the photon as somehow associated with a short-duration oscillatory wave pulse.



The atoms can be thought as emitting wavetrains that have a sustained phase for up to about 10 ns, after that a new wavetrain may be emitted with a totally random phase.

Flux density:

$$I = \langle E_0^2 \rangle = \sum_{i=1}^N \langle E_{0i}^2 \rangle + 2 \sum_{j>i}^N \sum_{i=1}^N E_{0i} E_{0j} \langle \cos(\alpha_i - \alpha_j) \rangle = \sum_{i=1}^N \langle E_{0i}^2 \rangle = N E_{01}^2$$

The resultant flux density arising from N sources having random, rapidly varying phases is given by N times the flux density of any one source.

Standing Waves

$$\frac{\partial^2}{\partial x^2}(\psi_1 + \psi_2) = \frac{1}{v^2} \frac{\partial^2}{\partial t^2}(\psi_1 + \psi_2)$$

Superposition principle

The general solution of the differential wave equation:

$$\varphi(x, t) = C_1 f(x - vt) + C_2 g(x + vt)$$

Examine two harmonic waves of the same frequency propagating in opposite direction:

An incoming wave traveling to the left: $E_I = E_{0I} \sin(kx + \omega t + \epsilon_I)$

It strikes a mirror at $x=0$, the reflected wave: $E_R = E_{0R} \sin(kx - \omega t + \epsilon_R)$

The composite wave: $E = E_I + E_R$

The two waves exist simultaneously in the region between the source and the mirror.

Assume that $E_{0I} = E_{0R}$; boundary conditions: $x = 0, E = 0$; set $\epsilon_I = 0$, so $\epsilon_R = 0$.

Composite disturbance

$$E = E_{0I} [\sin(kx + \omega t) + \sin(kx - \omega t)] = 2E_{0I} \sin kx \cos \omega t$$

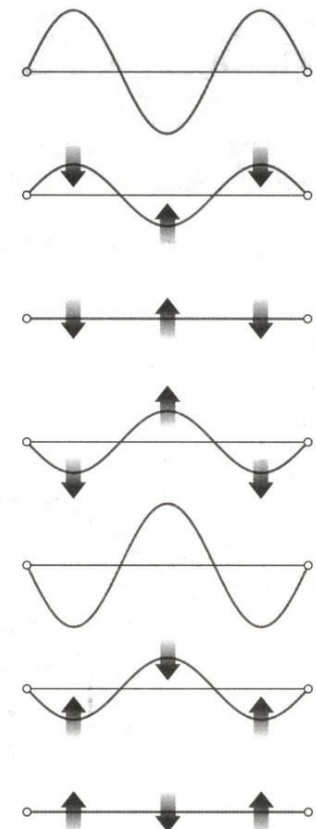
$$\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta)$$

Standing or stationary wave:

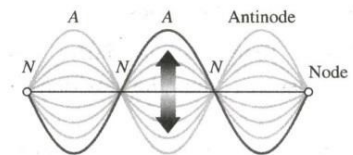
$$E = 2E_{0I} \sin kx \cos \omega t$$

Its profile does not move through space, because it is not of the form $f(x \pm vt)$

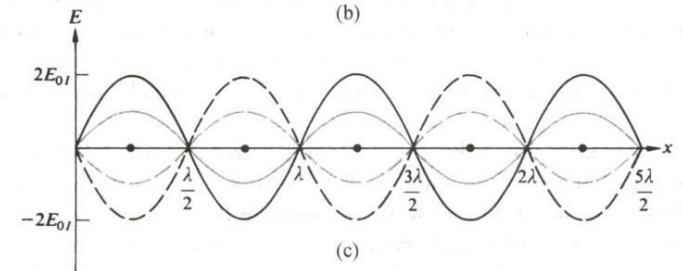
- ❑ At certain points (**nodes**—波节), $x=0, \lambda/2, \lambda, 3\lambda/2, \dots: E(x, t)=0$.
- ❑ Halfway between each adjacent node (**antinodes** —波腹): the amplitude has a maximum value of $\pm 2E_{0I}$.
- ❑ The disturbance $E(x, t)=0$, whenever $\cos \omega t=0$ or $t=(2m+1)\tau/4$, $m=0, 1, 2, \dots$



(a)



(b)



(c)

Standing wave water

Wiener's Experiment (Yr 1890)

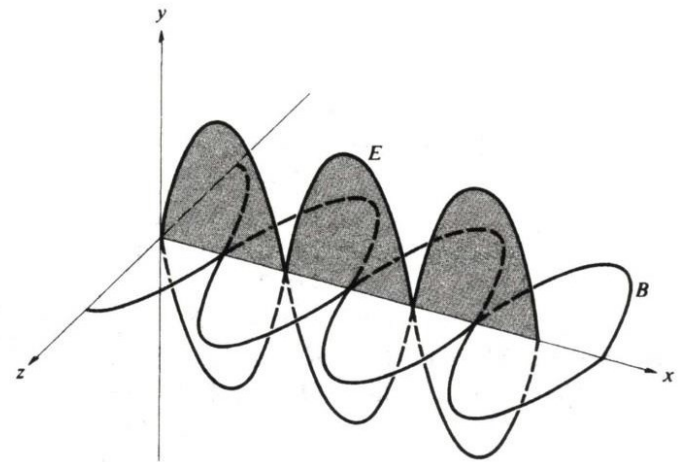
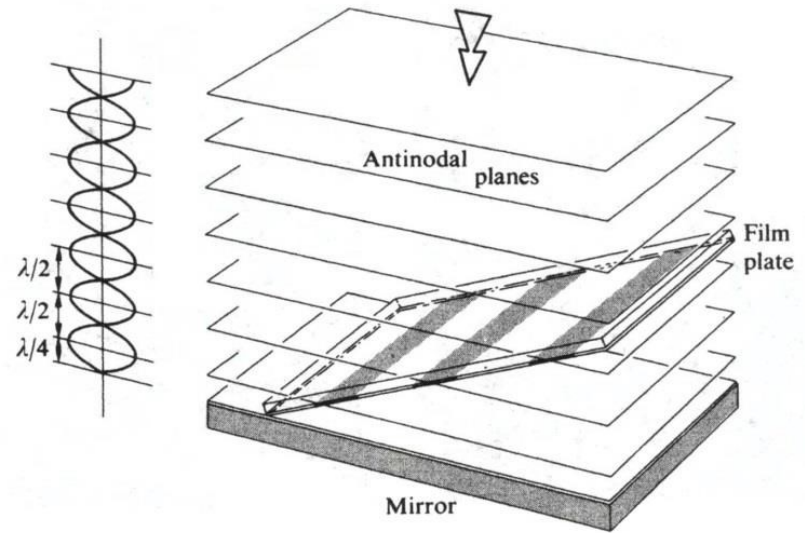
- ❑ Phenomena: the glass with emulsion is found to be blackened along a series of equidistant parallel bands.
- ❑ The black regions correspond to the place where the photographic layer had intersected the antinodal planes.

Energy conservation?

$$E(x, t) = 2E_{0I} \sin kx \cos \omega t$$

$$\begin{aligned} B(x, t) &= - \int \frac{\partial E}{\partial x} dt = -2E_0 k \cos kx \int \cos \omega t dt \\ &= -\frac{2E_0 k}{\omega} \cos kx \sin \omega t \\ &= -2B_0 \cos kx \sin \omega t \end{aligned}$$

It is the electric field that triggers the photochemical process.



The Addition of Waves of Different Frequency

Beats

$$\cos\alpha + \cos\beta = 2 \cos\frac{1}{2}(\alpha + \beta)\cos\frac{1}{2}(\alpha - \beta)$$

In the case of two waves of different frequency traveling in the same direction:

$$E_1 = E_{01}\cos(k_1x - \omega_1t) \quad \text{and} \quad E_2 = E_{01}\cos(k_2x - \omega_2t)$$

The net composite wave:

$$\begin{aligned} E &= E_{01}[\cos(k_1x - \omega_1t) + \cos(k_2x - \omega_2t)] \\ &= 2E_{01}\cos\frac{1}{2}[(k_1+k_2)x - (\omega_1 + \omega_2)t]\cos\frac{1}{2}[(k_1-k_2)x - (\omega_1 - \omega_2)t] \end{aligned}$$

Average angular frequency ($\bar{\omega}$)

$$\bar{\omega} \equiv \frac{1}{2}(\omega_1 + \omega_2)$$

Average propagation number ($\bar{k}$)

$$\bar{k} \equiv \frac{1}{2}(k_1 + k_2)$$

Modulation frequency (ω_m)

$$\omega_m \equiv \frac{1}{2}(\omega_1 - \omega_2)$$

Modulation propagation number (k_m)

$$k_m \equiv \frac{1}{2}(k_1 - k_2)$$

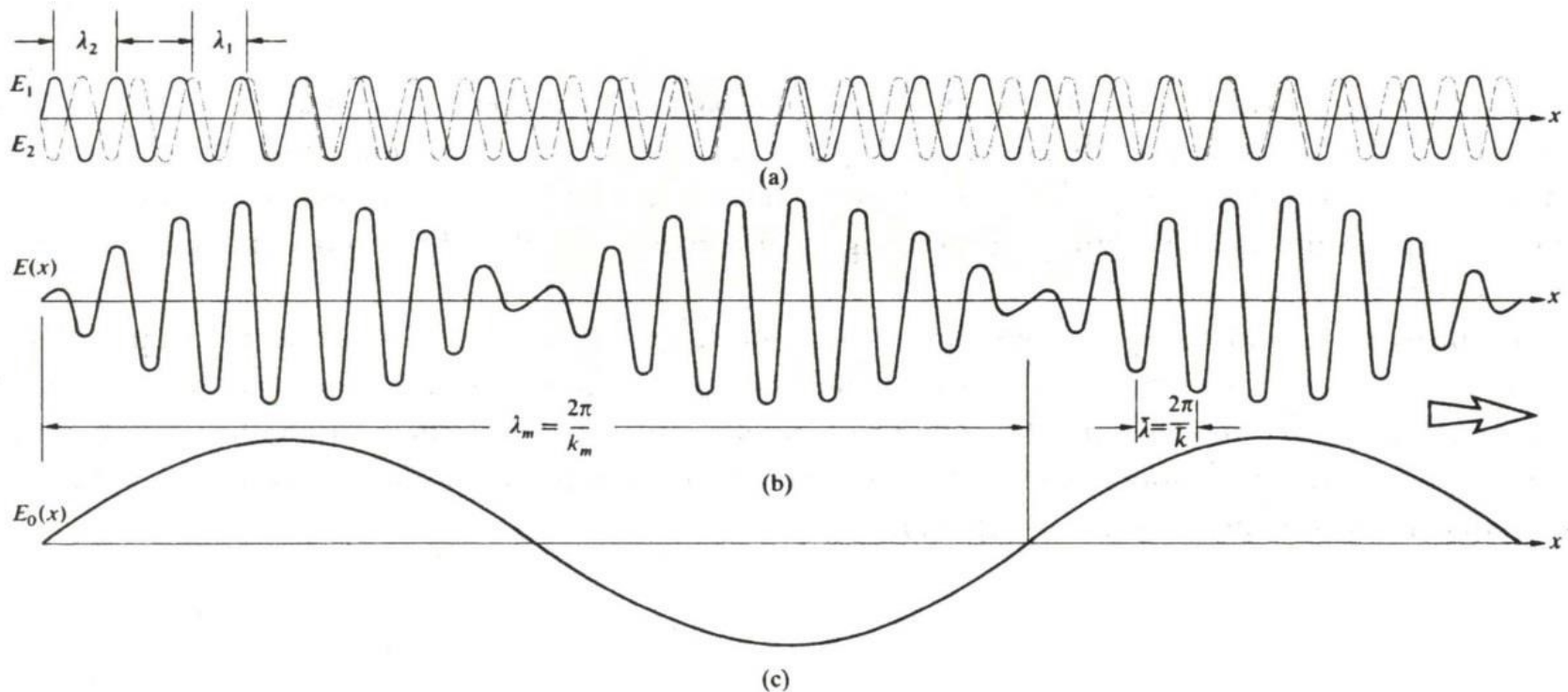
$$E(x, t) = 2E_{01} \cos(k_m x - \omega_m t) \cos(\bar{k}x - \bar{\omega}t)$$

Or rewrite as $E(x, t) = E_0(x, t) \cos(\bar{k}x - \bar{\omega}t)$

where $E_0(x, t) = 2E_{01} \cos(k_m x - \omega_m t)$

The total disturbance may be regarded as a traveling wave of frequency $\bar{\omega}$ having a time-varying or modulated amplitude $E_0(x, t)$.

For quasi-monochromatic light, $\omega_1 \approx \omega_2$, then $\bar{\omega} \gg \omega_m$ and $E_0(x, t)$ will change slowly.

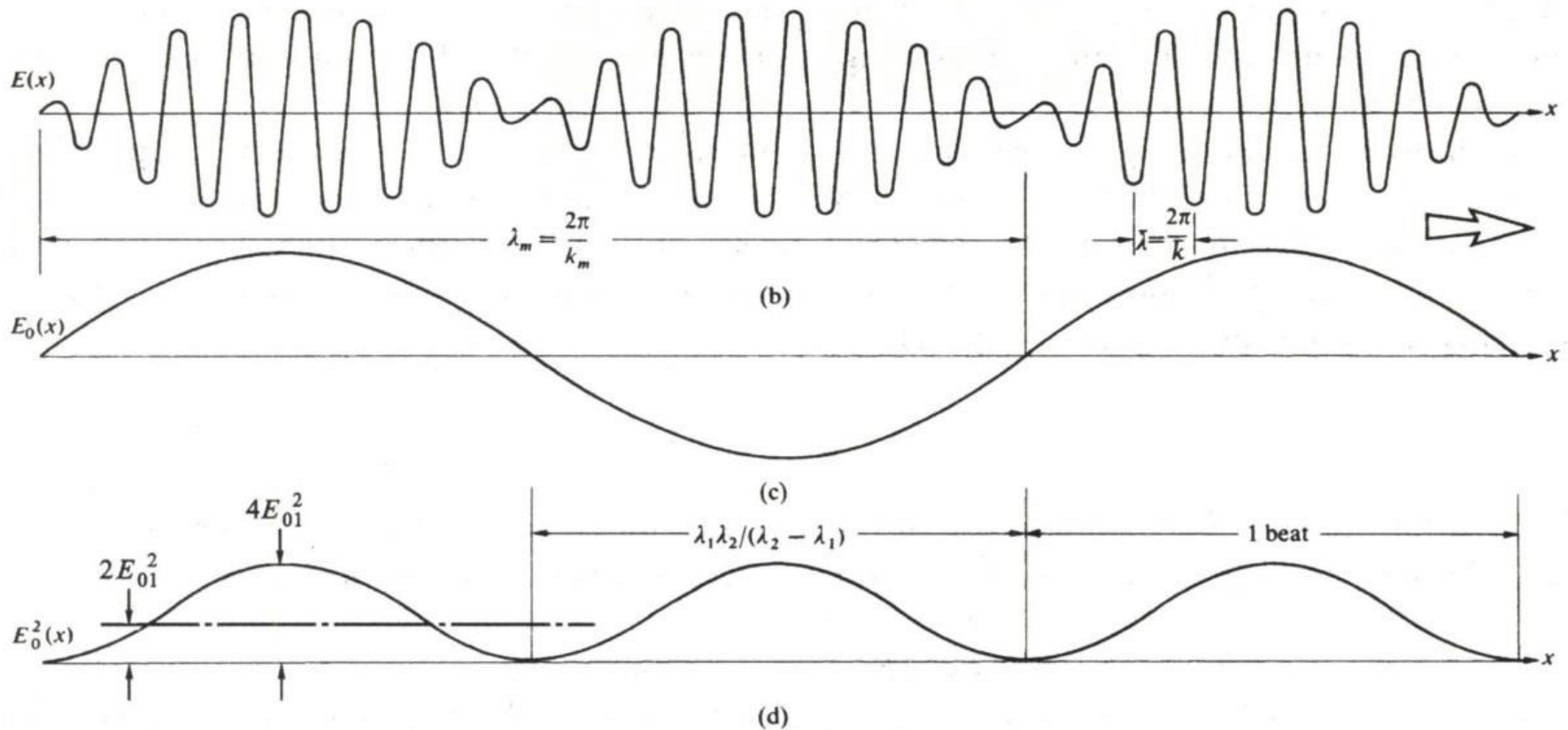


The irradiance:

$$E_0^2(x, t) = 4E_{01}^2(x, t)\cos^2(k_m x - \omega_m t)$$

$$= 2E_{01}^2(x, t)[1 + \cos(2k_m x - 2\omega_m t)]$$

Beat frequency: $2\omega_m$ or $(\omega_1 - \omega_2)$

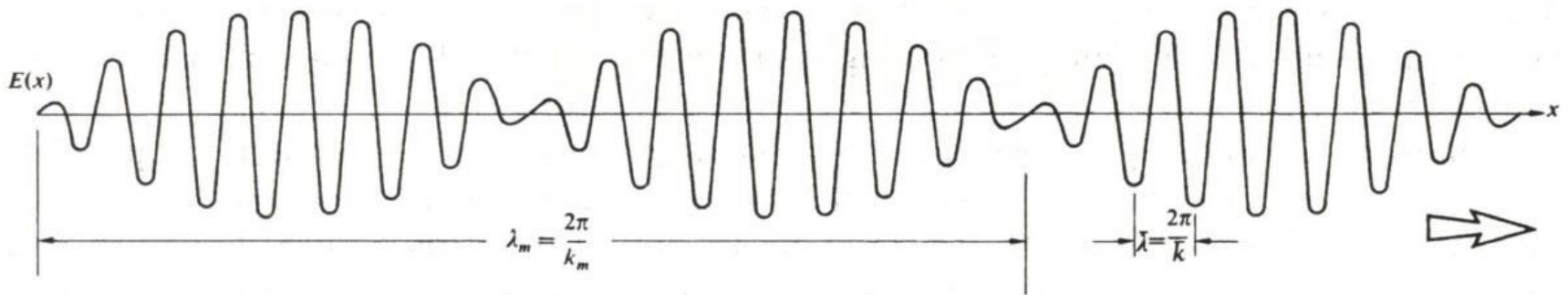


Velocity

The disturbance: $E(x, t) = E_0(x, t)\cos(\bar{k}x - \bar{\omega}t)$

It consists of a high-frequency ($\bar{\omega}$) carrier wave, amplitude-modulated by a cosine function.

Phase velocity (v): $v = -\frac{(\partial\varphi/\partial t)_x}{(\partial\varphi/\partial x)_t} = \frac{\bar{\omega}}{\bar{k}}$ it is independent of modulation



Group velocity (v_g): the rate at which the modulation envelope advances.

Suppose that the constituent waves, $E_1(x, t)$ and $E_2(x, t)$, advance with the same speed, $v_1 = v_2$. And the two harmonic functions having different wavelengths and frequencies. $v_g = v = v_1 = v_2$

The expression for the modulation envelope:

$$E_0(x, t) = 2E_{01}(x, t)\cos(k_mx - \omega_mt)$$

The modulation advances at a rate (**group velocity**) dependent on the phase of the envelope $(k_mx - \omega_mt)$

$$v_g = -\frac{(\partial\varphi/\partial t)_x}{(\partial\varphi/\partial x)_t} = \frac{\omega_m}{k_m} = \frac{\omega_1 - \omega_2}{k_1 - k_2} = \frac{\Delta\omega}{\Delta k}$$

In a dispersive media, ω is dependent on λ , or equivalently on k .

When frequency range $\Delta\omega$ is small, centered about $\bar{\omega}$, the **group velocity** equals to:

$$v_g = \frac{d\omega}{dk}$$

$$v_1 = \frac{\omega_1}{k_1} = \frac{c}{n(\lambda_1)} \quad v_2 = \frac{\omega_2}{k_2} = \frac{c}{n(\lambda_2)}$$

$$\begin{aligned} v_g = \frac{\Delta\omega}{\Delta k} &= \frac{v_1 k_1 - v_2 k_2}{k_1 - k_2} = \textcolor{red}{v_1} + \frac{k_2}{k_1 - k_2} (v_1 - v_2) \\ &= \textcolor{red}{v_2} + \frac{k_1}{k_1 - k_2} (v_1 - v_2) \end{aligned}$$

$$v_g = \bar{v} + \bar{k} \cdot \frac{\Delta v}{\Delta k}$$

Average velocity

$$\bar{v} = \frac{1}{2} (v_1 + v_2)$$

Average propagation number

$$\bar{k} = \frac{1}{2} (k_1 + k_2)$$

Dispersive term

$$\frac{\Delta v}{\Delta k} = \frac{v_1 - v_2}{k_1 - k_2}$$

$$v_g = \frac{d\omega}{dk} \qquad \omega = \frac{kc}{n}$$

$$v_g = \frac{d\omega}{dk} = \frac{c}{n} - \frac{kc}{n^2} \frac{dn}{dk} = v \left(1 - \frac{k}{n} \frac{dn}{dk} \right)$$

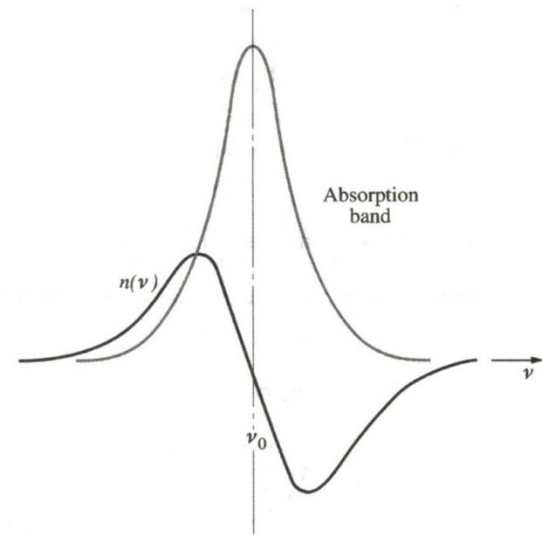
Normal dispersion $v_g < \bar{v}$

No dispersion $v_g = \bar{v}$

Abnormal dispersion $v_g > \bar{v}$

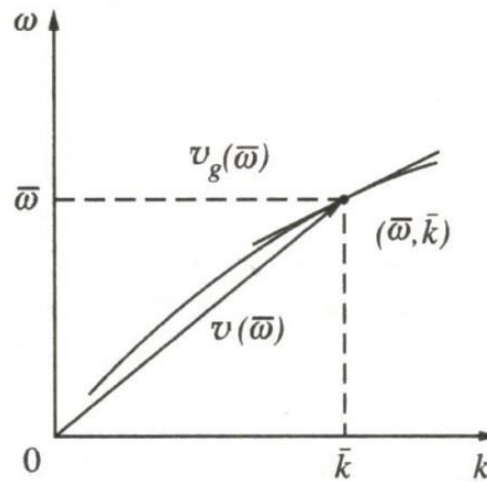
□ $n(\nu)$ curve (typical medium)

- ✓ Normal dispersion: $n(\nu)$ increases with increasing ν .
- ✓ Anomalous dispersion: $n(\nu)$ decreases with increasing ν .



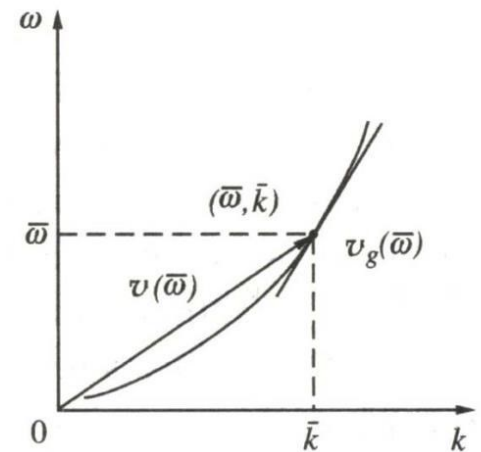
A typical representation of the frequency dependence of the index of refraction in the vicinity of an atomic resonance.

□ Dispersion relation:
convex upward for
normal dispersion
and concave
downward for
anomalous
dispersion



(a)

normal dispersion



(b)

anomalous dispersion

Since $\omega = kv$

From $v_g = \frac{d\omega}{dk}$, we have $v_g = v + k \frac{dv}{dk}$

In nondispersive media: v is independent of λ .

So $v_g = v$

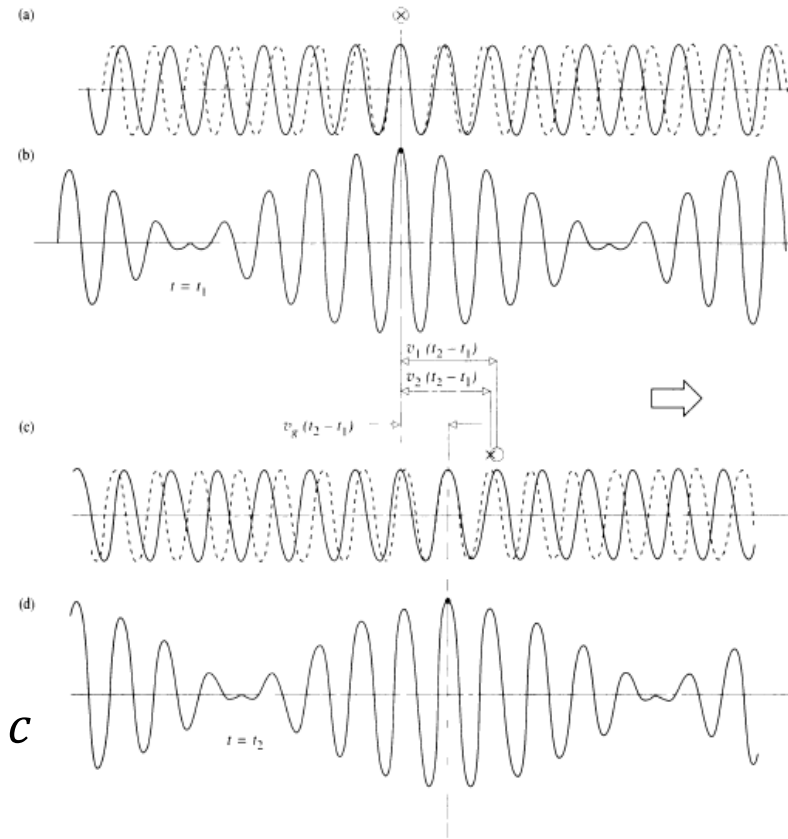
Specifically, in vacuum $\omega = kc$, $v = c$, and $v_g = c$

In dispersive media ($n(k)$): $\omega = \frac{kc}{n}$

So $v_g = \frac{c}{n} - \frac{kc}{n^2} \frac{dn}{dk}$, or $v \left(1 - \frac{k}{n} \frac{dn}{dk} \right)$

For optical media in regions of normal dispersion, $\frac{dn}{dk} > 0$, so $v_g < v$.

In regions of anomalous dispersion, $\frac{dn}{dk} < 0$, so $v_g > v$.

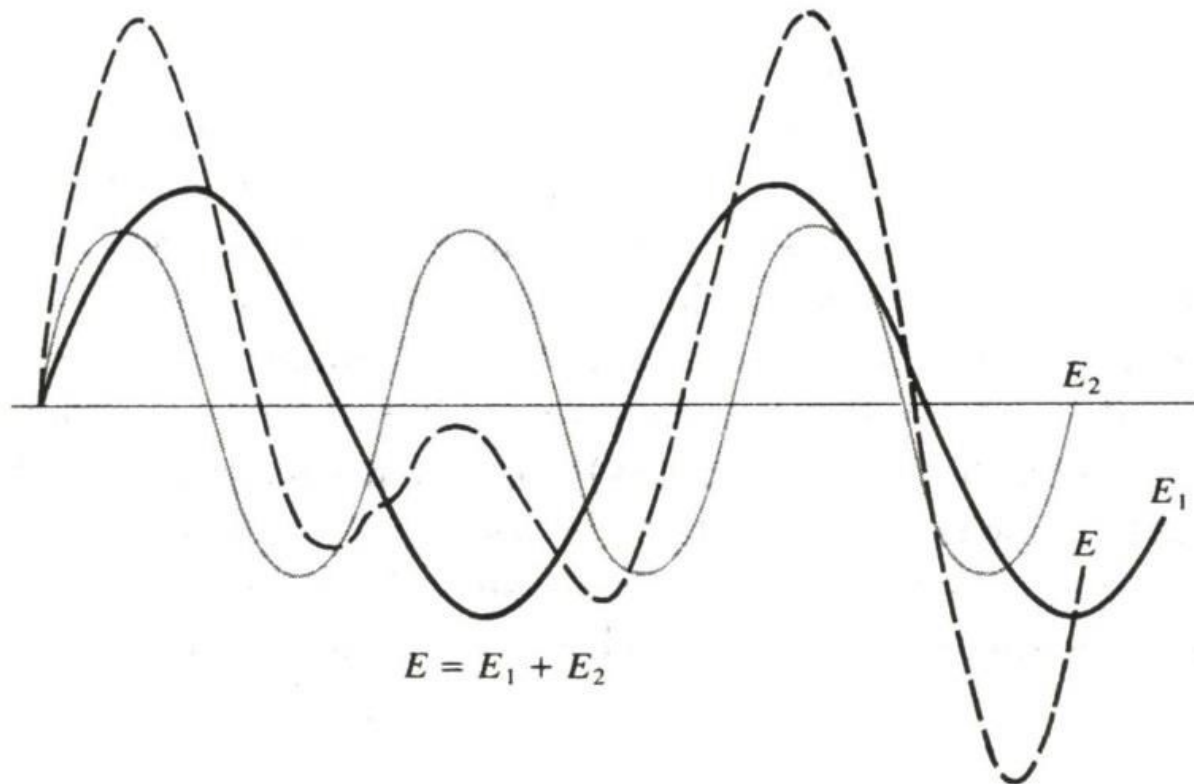


Homework

□ 7.1, 7.5, 7.19, 7.24

6.3 Anharmonic Periodic Waves-Fourier Analysis

The composite disturbance is **anharmonic** (or it is not sinusoidal), when two harmonic functions having **different amplitudes and wavelengths** superpose.



Fourier's theorem:

A function $f(x)$, having a spatial period λ , can be synthesized by a sum of harmonic functions whose wavelengths are integral submultiples of λ (that is, λ , $\lambda/2$, $\lambda/3$, etc.)

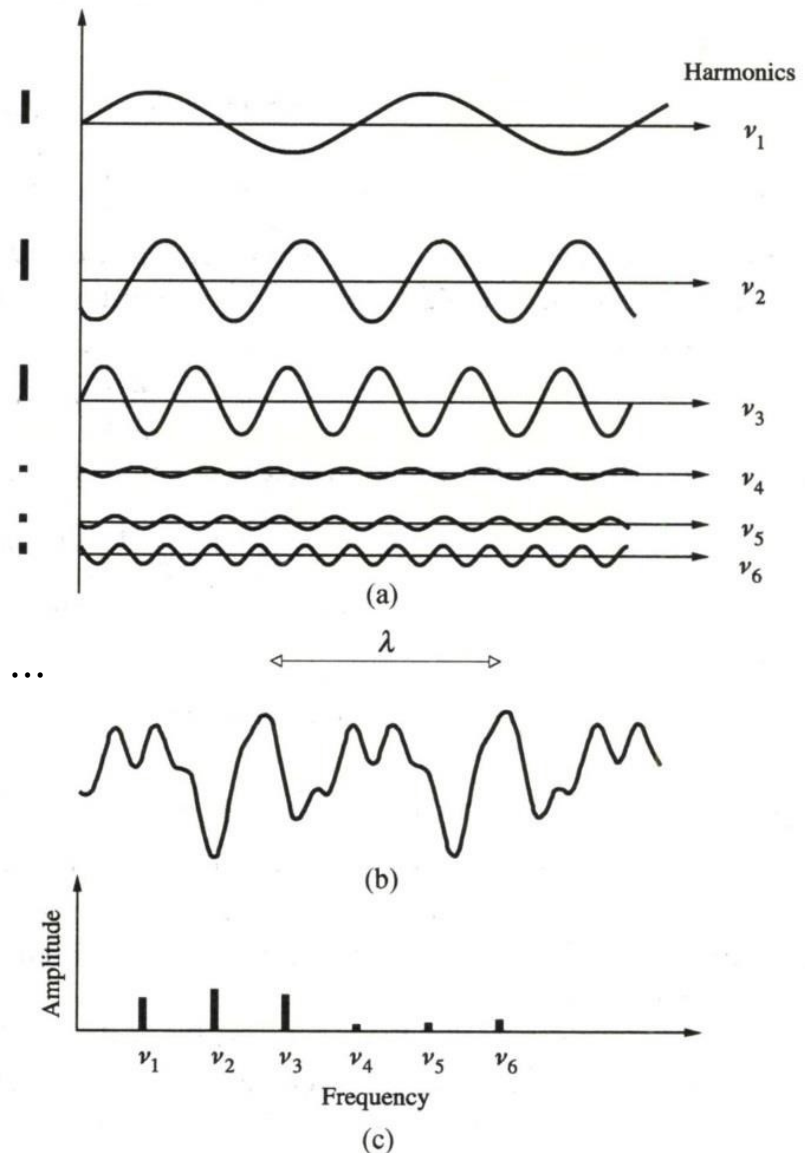
The Fourier-series representation:

$$f(x) = C_0 + C_1 \cos\left(\frac{2\pi}{\lambda}x + \epsilon_1\right) + C_2 \cos\left(\frac{2\pi}{\lambda/2}x + \epsilon_2\right) + \dots$$

where the C -value are constants

The profile $f(x)$ may correspond to a traveling wave $f(x - vt)$.

The synthesized function comprises an infinite number of terms, the series will presumably be identical to $f(x)$



The Fourier-series representation:

$$f(x) = C_0 + C_1 \cos\left(\frac{2\pi}{\lambda}x + \epsilon_1\right) + C_2 \cos\left(\frac{2\pi}{\lambda/2}x + \epsilon_2\right) + \dots$$

$$C_m \cos(mkx + \epsilon_m) = A_m \cos mkx + B_m \sin mkx$$

where $k = \frac{2\pi}{\lambda}$, $A_m = C_m \cos \epsilon_m$, $B_m = -C_m \sin \epsilon_m$

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cos mkx + \sum_{m=1}^{\infty} B_m \sin mkx$$

The process of determining the coefficients A_0 , A_m , and B_m for a specific periodic function $f(x)$ is referred to **Fourier Analysis**.

$$\int_0^\lambda \sin mkx \, dx = \int_0^\lambda \cos mkx \, dx = 0$$

$$\int_0^\lambda f(x) \, dx = \int_0^\lambda \frac{A_0}{2} \, dx = A_0 \frac{\lambda}{2}$$

$$A_0 = \frac{2}{\lambda} \int_0^\lambda f(x) \, dx$$

$$\int_0^\lambda \sin akx \cos bkx \, dx = 0$$

$$\int_0^\lambda \cos akx \cos bkx \, dx = \frac{\lambda}{2} \delta_{ab}$$

$$\int_0^\lambda \sin akx \sin bkx \, dx = \frac{\lambda}{2} \delta_{ab}$$

$$\int_0^\lambda f(x) \cos mkx \, dx = \int_0^\lambda A_m \cos^2 mkx \, dx = \frac{\lambda}{2} A_m$$

$$A_m = \frac{2}{\lambda} \int_0^\lambda f(x) \cos mkx \, dx$$

$$B_m = \frac{2}{\lambda} \int_0^\lambda f(x) \sin mkx \, dx$$

A periodic function $f(x)$ can be represented as a Fourier series

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cos mkx + \sum_{m=1}^{\infty} B_m \sin mkx$$

Knowing the function $f(x)$, the coefficients A_m and B_m can be computed using the following formula

$$A_0 = \frac{2}{\lambda} \int_0^{\lambda} f(x) dx$$

$$A_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \cos mkx dx$$

$$B_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \sin mkx dx$$

For even function $f(x)$

$$f(-x) = f(x); B_m = 0 \text{ for all } m.$$

For odd function $f(x)$

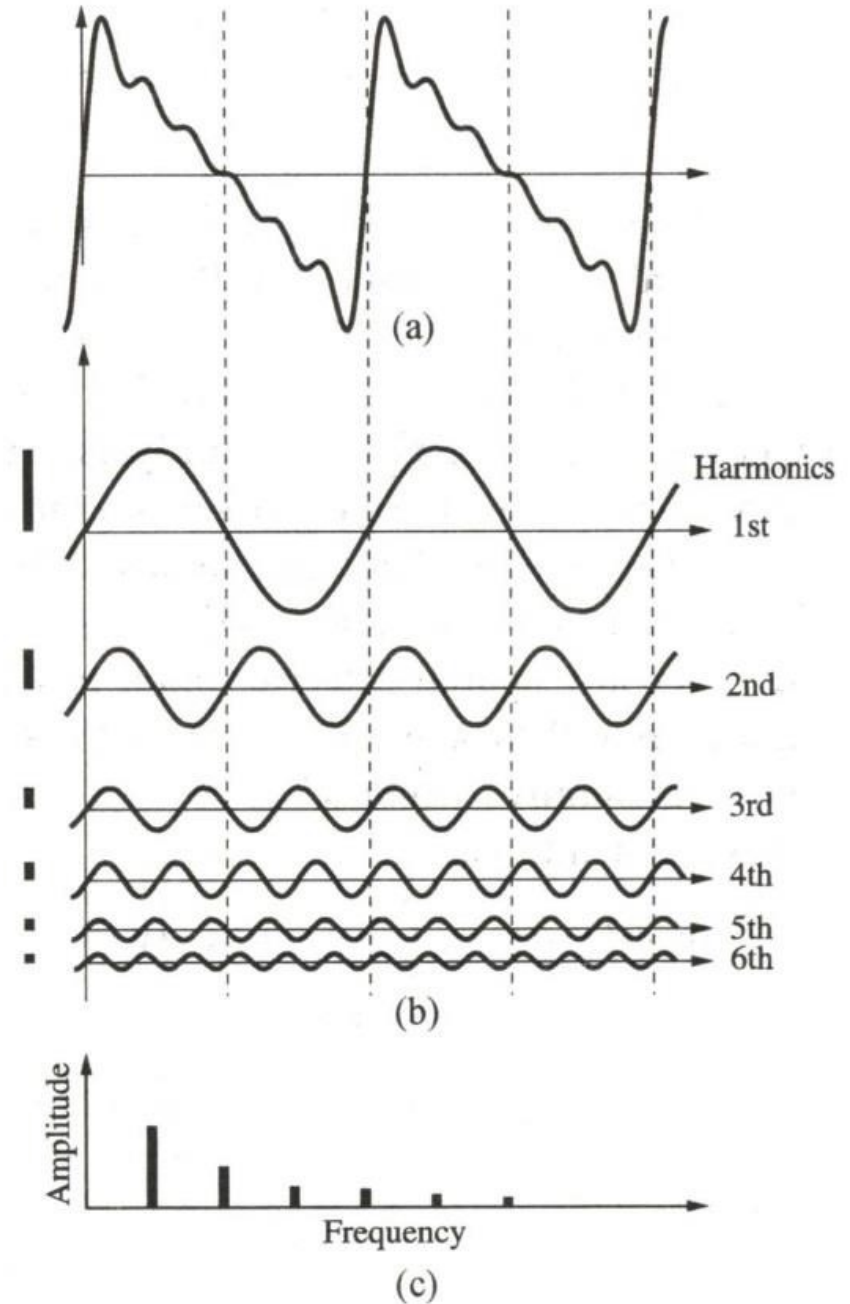
$$f(-x) = -f(x); A_m = 0 \text{ for all } m.$$

Example: the saw-tooth function

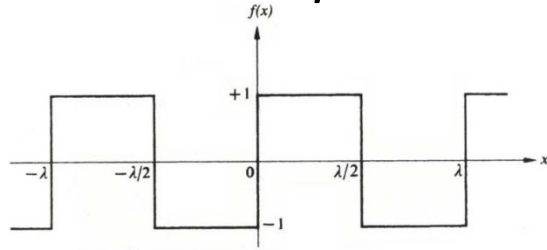
$f(-x) = -f(x)$; so $A_m = 0$ for all m .

The Fourier-series representation:

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} B_m \sin mkx$$



Example: compute the Fourier series that corresponds to a *square wave*.



$$f(x) = \begin{cases} +1 & \text{when } 0 < x < \frac{\lambda}{2} \\ -1 & \text{when } \frac{\lambda}{2} < x < \lambda \end{cases}$$

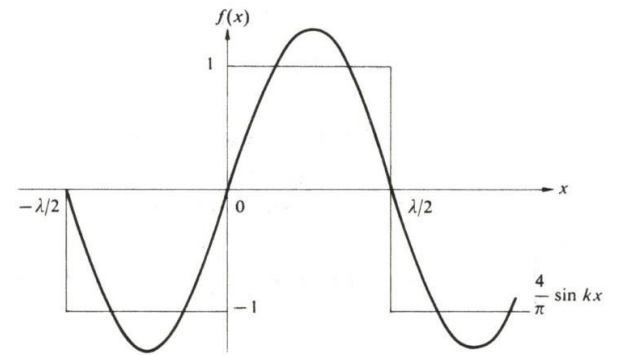
$$B_m = \frac{2}{\lambda} \int_0^{\lambda/2} (+1) \sin mkx \, dx + \frac{2}{\lambda} \int_{\lambda/2}^{\lambda} (-1) \sin mkx \, dx$$

$$B_m = \frac{1}{m\pi} [-\cos mkx]_0^{\lambda/2} + \frac{1}{m\pi} [\cos mkx]_{\lambda/2}^{\lambda}$$

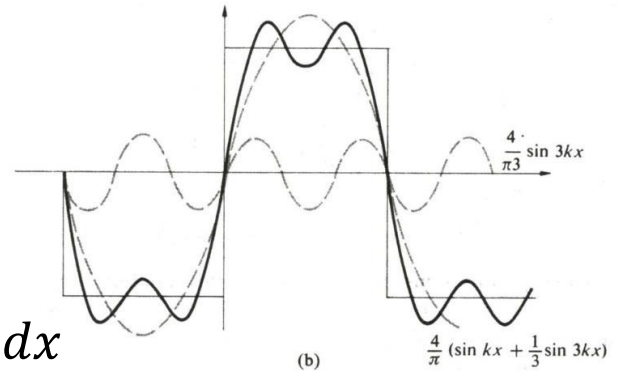
$$B_1 = \frac{4}{\pi}, \quad B_2 = 0, \quad B_3 = \frac{4}{3\pi}, \quad B_4 = 0, \quad B_5 = \frac{4}{5\pi}, \dots$$

The required series:

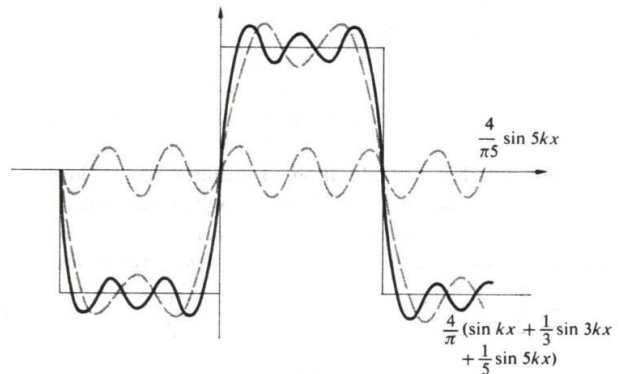
$$f(x) = \frac{4}{\pi} \left(\sin kx + \frac{1}{3} \sin 3kx + \frac{1}{5} \sin 5kx + \dots \right)$$



(a)



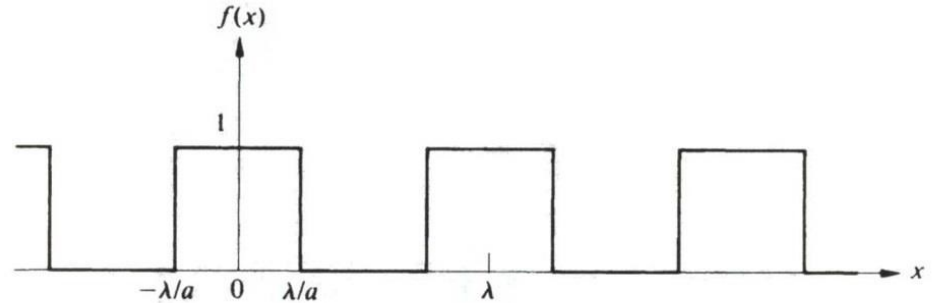
(b)



(c)

Example II: a square wave.

$$f(x) = \begin{cases} 1 & \text{when } -\frac{\lambda}{a} < x < \frac{\lambda}{a} \\ 0 & \text{others} \end{cases}$$



The Fourier series:

$$f(x) = \frac{2}{a} + \sum_{m=1}^{\infty} \frac{4}{a} \text{sinc}(m2\pi/a) \cos mkx$$

where $\text{sinc}(m2\pi/a) = \frac{\text{sinc}(m2\pi/a)}{m2\pi/a}$

The spatial period: λ ;

The spatial frequency: $\kappa=1/\lambda$;

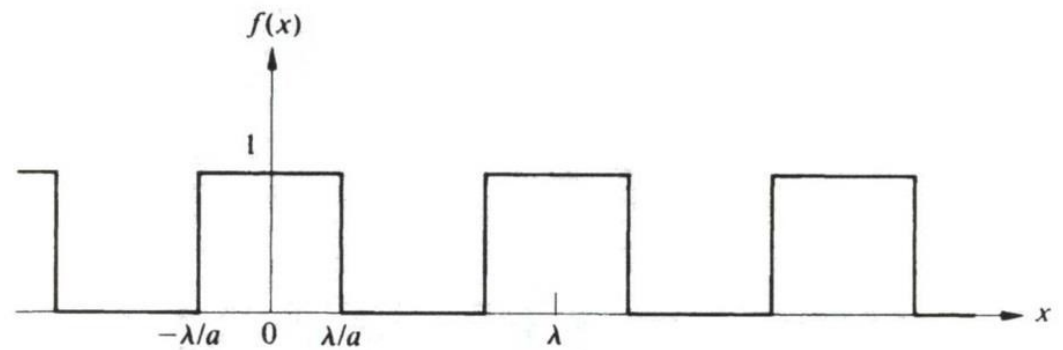
The angular spatial frequency: $k=2\pi\kappa$

Note: symbols k and λ for lightwave itself, $\mathbf{k}$ and $\boldsymbol{\lambda}$ to describe spatial optical patterns.

$$A_m = \frac{4}{a} \left(\frac{\sin m2\pi/a}{m2\pi/a} \right)$$

$$m = 1, 2, \dots$$

$$\text{So } A_1 = \frac{4}{a} \left(\frac{\sin 2\pi/a}{2\pi/a} \right)$$



$$\text{The ratio: } \frac{A_m}{A_1} = \frac{\sin m2\pi/a}{m\sin 2\pi/a}$$

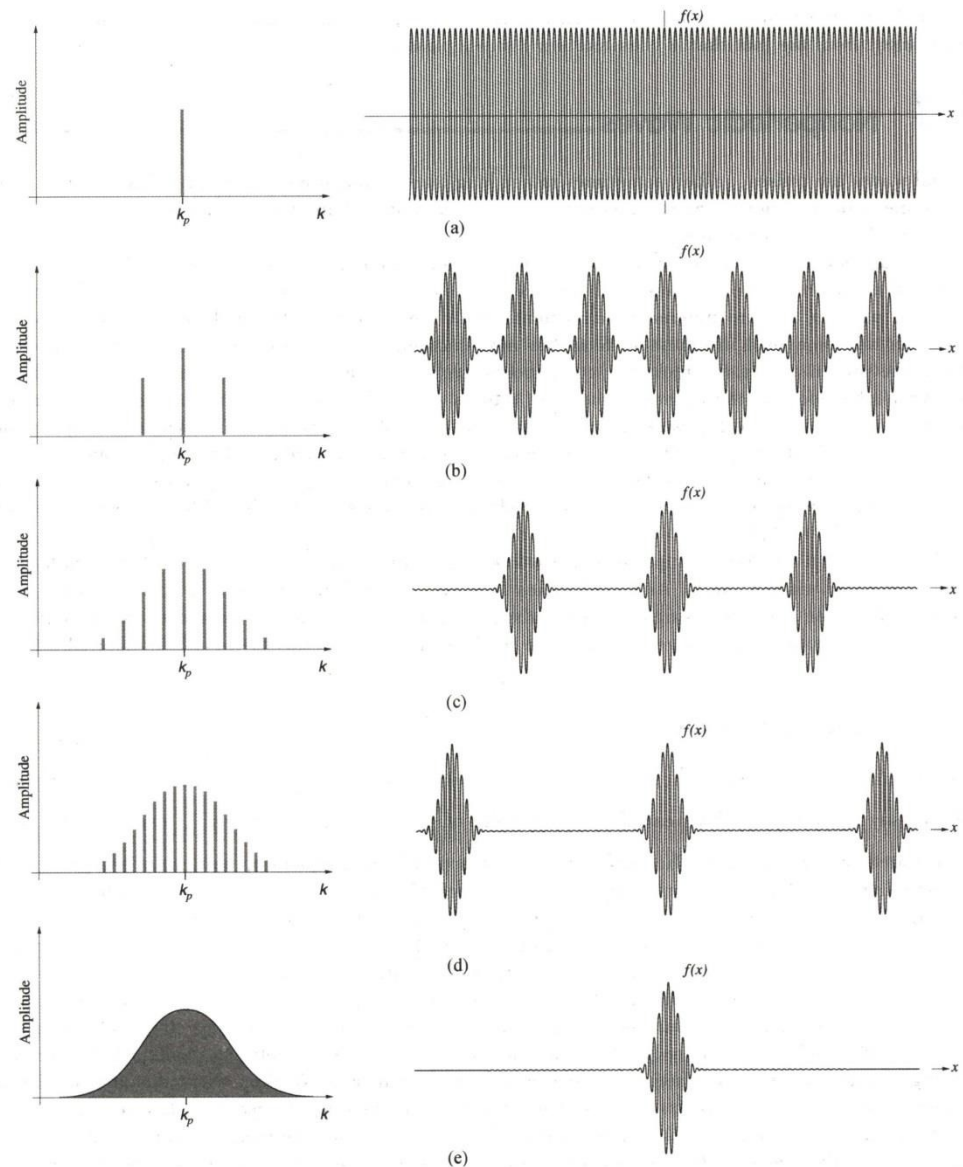
$$\text{For } a=4; m=9: A_9 \approx 10\% A_1$$

$$\text{For } a=400; m=9: A_9 \approx 99\% A_1$$

Making the peak narrower has the effect of introducing higher-order harmonics, which in turn have smaller wavelengths.

6.4 Nonperiodic Waves-Fourier Integrals

All real waves are pulses, albeit sometimes rather long ones, and so it is important to learn how to analyze nonperiodic functions.



Fourier integrals

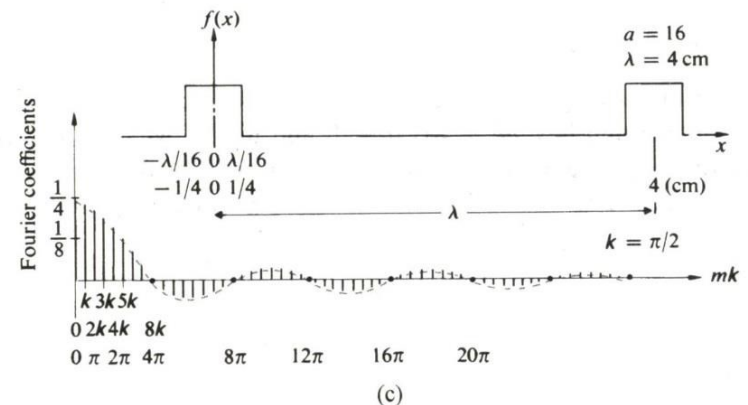
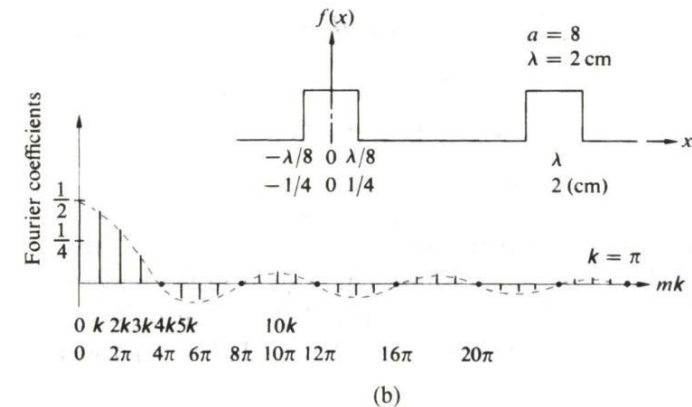
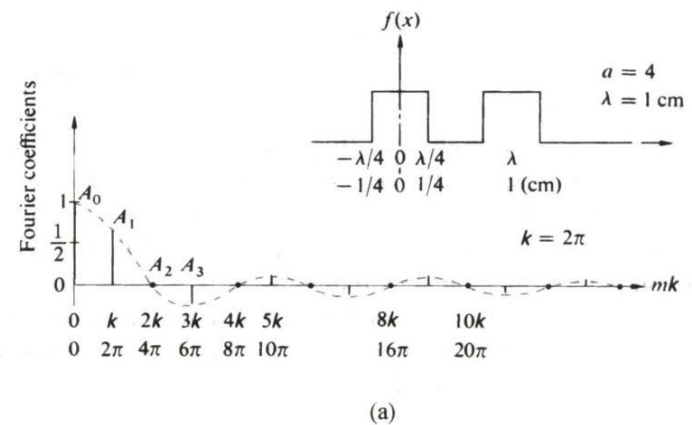
$$A_m = \frac{4}{a} \left(\frac{\sin m2\pi/a}{m2\pi/a} \right)$$

$$f(x) = \frac{2}{a} + \sum_{m=1}^{\infty} \frac{4}{a} \operatorname{sinc} m2\pi/a \cos mkx$$

The importance of each particular frequency, mk , can be appreciated by examining the value of the corresponding Fourier coefficient, in this case A_m , which is thought of as the weighting factors that appropriately emphasize the various harmonics.

As λ increases and the function takes on the appearance of a single square pulse, the space between each of the $A(mk)$ contributions in the spectrum will decrease.

In the limit as λ approaches ∞ , the spectral lines will become infinitely close to each other.



As λ goes to infinity, Fourier integral changes from:

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cos mkx + \sum_{m=1}^{\infty} B_m \sin mkx$$

$$A_0 = \frac{2}{\lambda} \int_0^{\lambda} f(x) dx \quad A_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \cos mkx dx \quad B_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \sin mkx dx$$

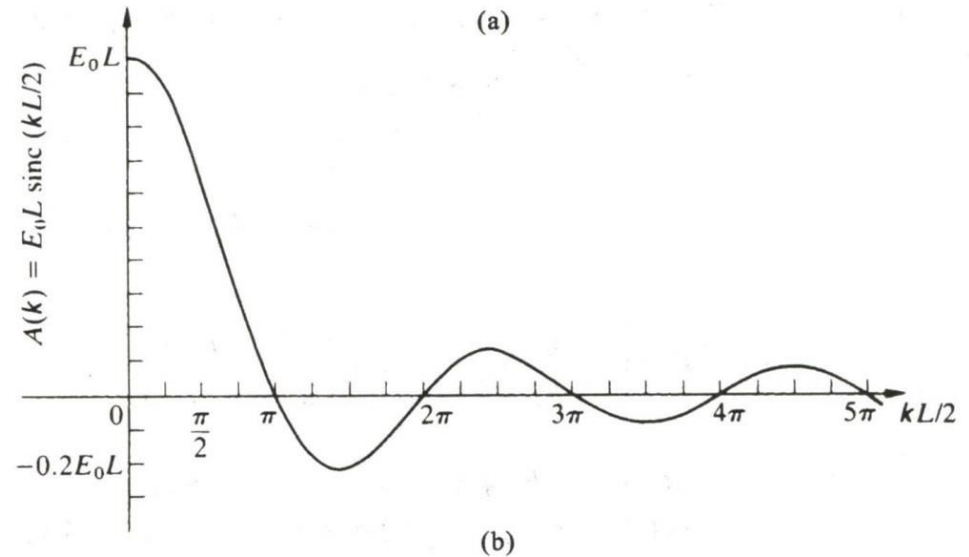
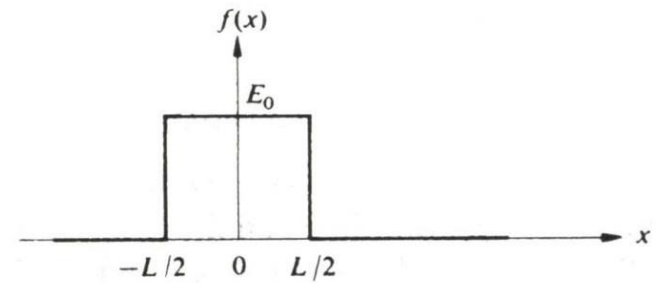
to

$$f(x) = \frac{1}{\pi} \left[\int_0^{\infty} A(k) \cos kx dk + \int_0^{\infty} B(k) \sin kx dk \right]$$

$$A(k) = \int_{-\infty}^{\infty} f(x) \cos kx dx \quad B(k) = \int_{-\infty}^{\infty} f(x) \sin kx dx$$

Pulses and Wave Packets

$$f(x) = \begin{cases} E_0 & \text{when } |x| < \frac{L}{2} \\ 0 & \text{when } |x| > \frac{L}{2} \end{cases}$$

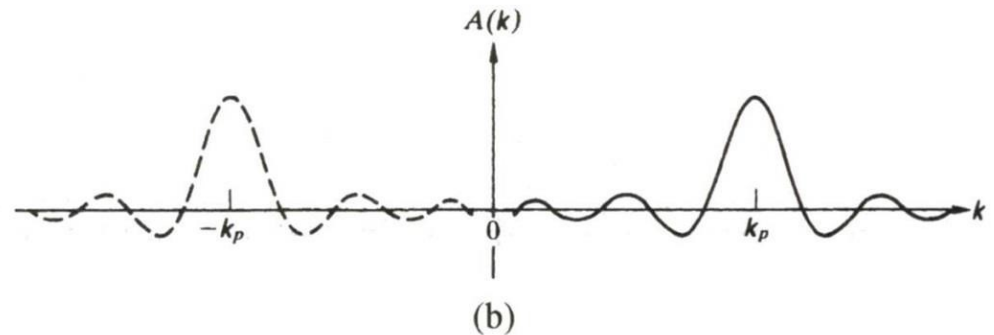
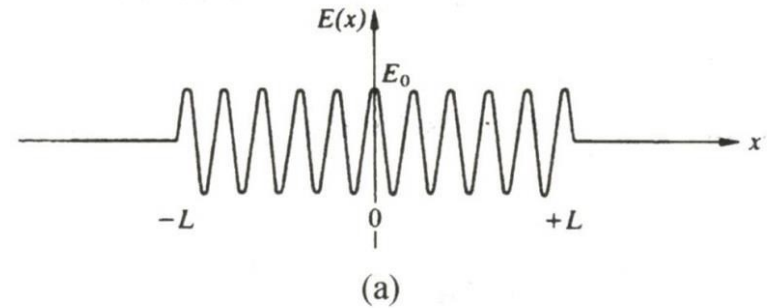


$$A(k) = E_0 L \left(\frac{\sin \frac{kL}{2}}{\frac{kL}{2}} \right) = E_0 L \text{sinc } kL/2$$

The cosine wavetrain:

$$E(x) = \begin{cases} E_0 \cos k_p x & \text{when } -L \leq x \leq L \\ 0 & \text{when } |x| > L \end{cases}$$

k_p : the spatial frequency of the harmonic region of the pulse itself.

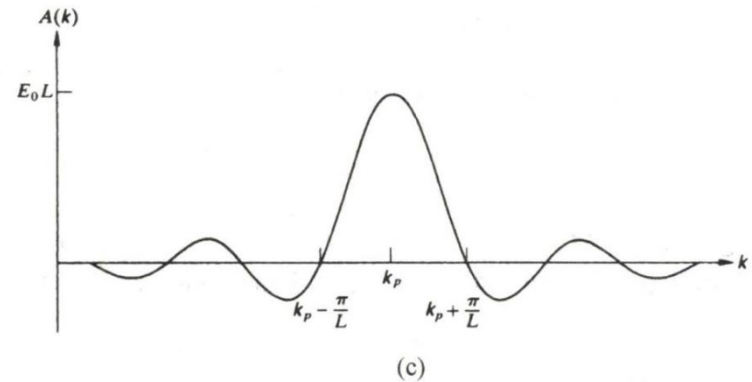


$$A(k) = E_0 L \left[\frac{\sin(k_p + k)L}{(k_p + k)L} + \frac{\sin(k_p - k)L}{(k_p - k)L} \right]$$

For the case of many waves in the train ($\lambda_p \ll L$)

$$k_p L \gg 2\pi \quad (k_p + k)L \gg 2\pi$$

$\text{sinc}(k_p + k)L$ is a fairly small value.



Limit the treatment to the only positive values

$$k_p = k \quad \frac{\sin(k_p - k)L}{(k_p - k)L} = 1$$

$$A(k) = E_0 L \left[\frac{\sin(k_p + k)L}{(k_p + k)L} + \frac{\sin(k_p - k)L}{(k_p - k)L} \right]$$

$$A(k) = E_0 L \frac{\sin(k_p - k)L}{(k_p - k)L}$$

- It can be thought of as the composite of an infinite ensemble of harmonic waves, which is spoke **as wave packets** or **wave groups**.
- When $L \rightarrow \infty$, its frequency shrinks, and the curve closes down to a single tall spike at k_p (idealized monochromatic wave in the limit case).

The frequency bandwidths:

$$\Delta k = 2\pi/L$$

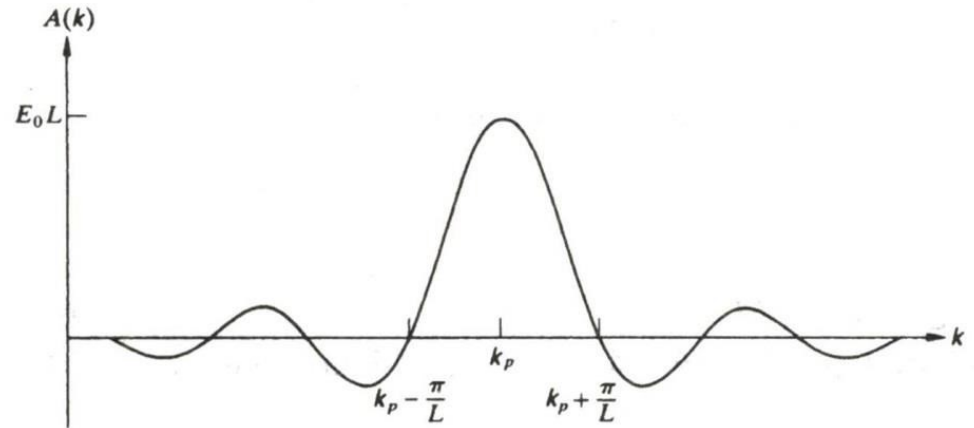
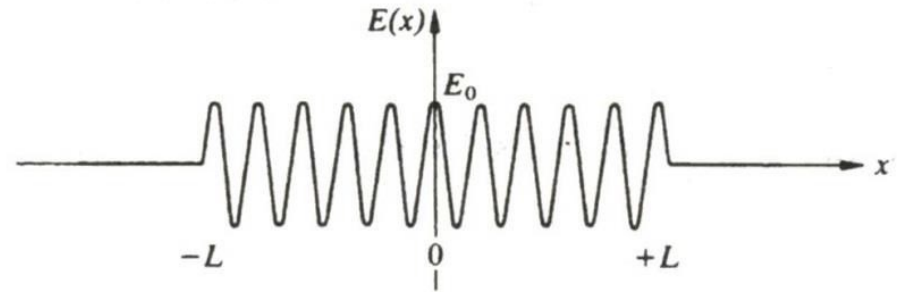
The spatial extent of the pulse:

$$\Delta x = 2L$$

The product:

$$\Delta k \Delta x = 4\pi$$

If the wave packet has a narrow bandwidth, it will extend over a large region of space and time.



The wave packet in the time domain:

$$E(t) = \begin{cases} E_0 \cos \omega_p t & \text{when } -T \leq t \leq T \\ 0 & \text{when } |t| > T \end{cases}$$

$$\Delta \omega = 2\pi/T$$

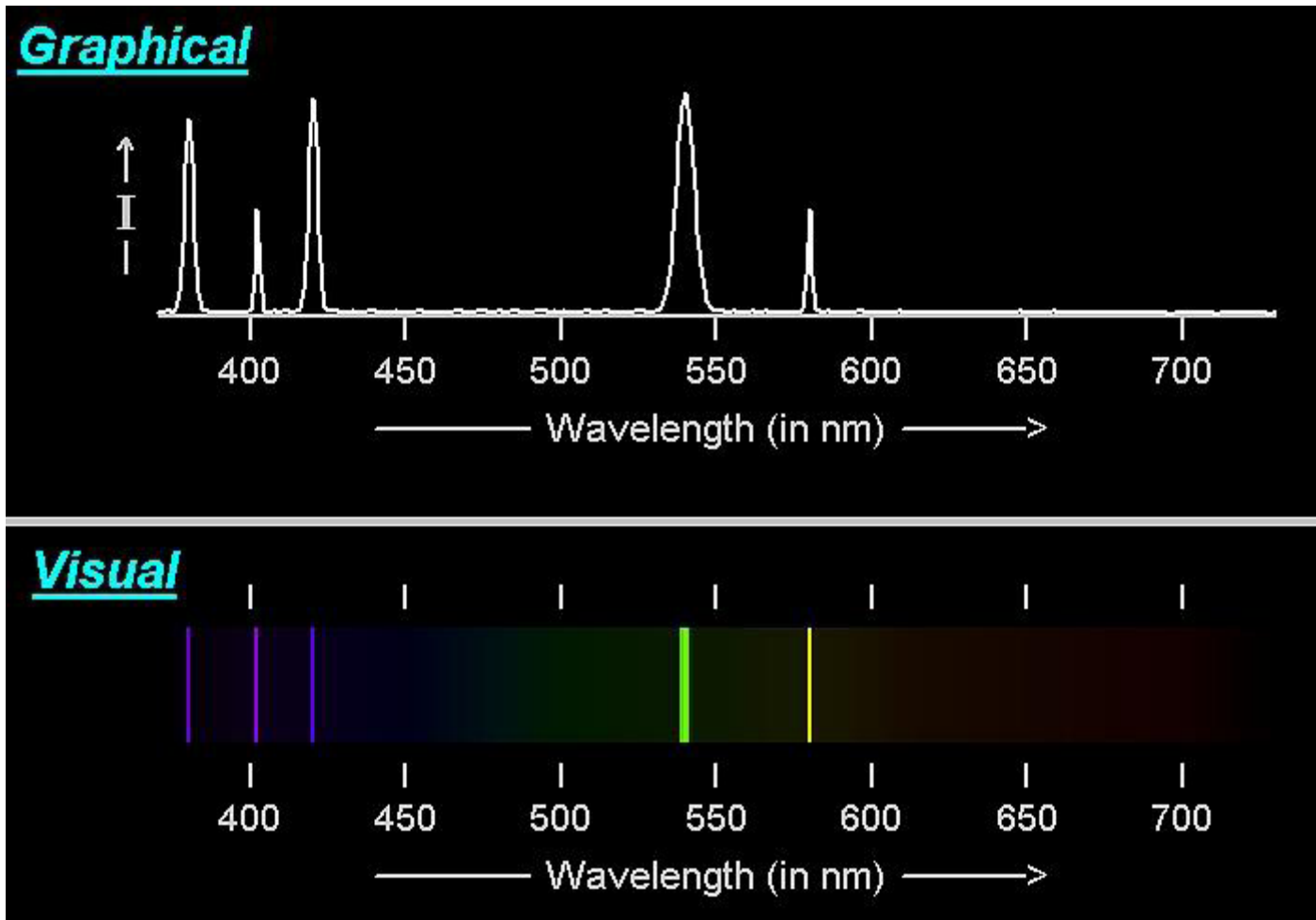
$$\Delta t = 2T$$

$$A(\omega) = E_0 T \operatorname{sinc}(\omega_p - \omega)T$$

$$\Delta t \Delta \omega = 4\pi$$

The k and ω are related by the phase velocity

Coherence Length



In any event, the individual spectral lines are never infinitely sharp, they always consist of a band of frequencies, however small.

Natural linewidth: The electron transitions responsible for the generation of light have a duration on the order of 10^{-8} or 10^{-9} s. Because the emitted wavetrains are finite, there will be a spread in the frequencies.

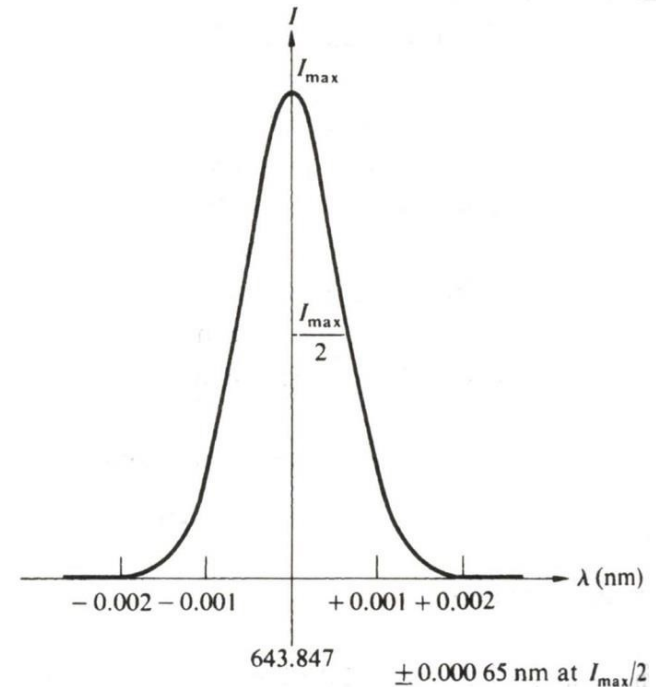
Other reason broadening the frequency distribution: **random thermal motion; collisions.**

Coherence time: $\Delta\nu \sim 1/\Delta t$

Coherence length: $\Delta x_c = c\Delta t_c$.

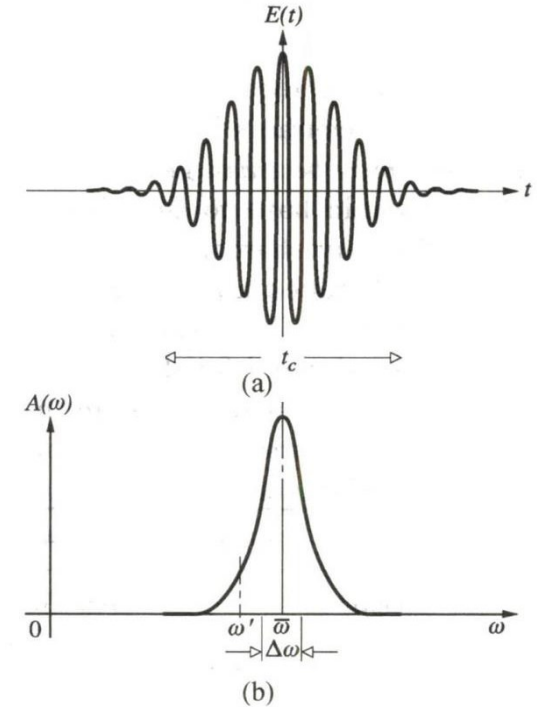
The coherence length is the extent in space over which the wave is nicely sinusoidal so that its phase can be predicted reliably.

Experimentally, the irradiance versus frequency is found to be Gaussian. So the net amplitude of the light field is also Gaussian.



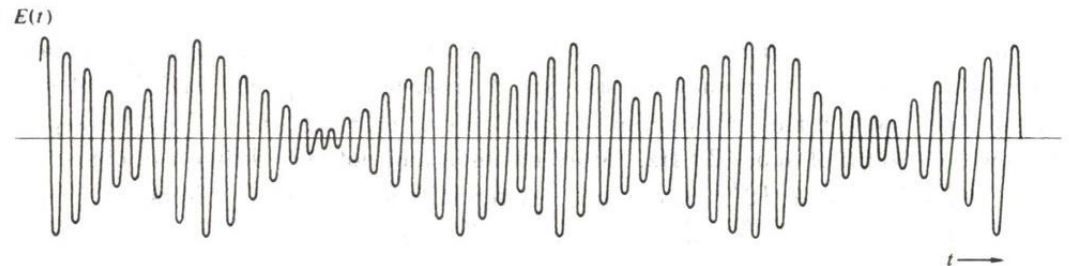
The frequency distribution of a spectral line from a quasimonochromatic source

- ❑ A single photon wavetrain: one of N identical such packets making up the beam.
- ❑ Consider the frequency component (ω'): an infinitely long, constant-amplitude sinusoid in a wavetrain; every packet has the identical amplitude and frequency.
- ❑ At any point of a stream of photons: the superposition of ω' component with random relative phase
- ❑ An average to a harmonic wave of frequency ω' with the amplitude proportional $N^{1/2}$.
- ❑ It is the same for every other frequency constituting the packets.



The observed spectral line (Gaussian function) corresponds to the power spectrum of the beam, it also corresponds to the power spectrum of an individual photon packet.

Different relative phases among wavetrains, so the composite lightwave can be pictured as following picture.



Homework

□ 7.32, 7.33, 7.37, 7.39

In Problem 7.33, the figure 7.20 should be 7.30